Electromagnetic Scattering Feature (ESF) Module Embedded Network Based on ASC Model for Robust and Interpretable SAR ATR

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***Abstract*— Deep learning has been widely used in automatic target recognition (ATR) for synthetic aperture radar (SAR) recently. However, most of the studies are based on the network structure in optical images and lack full consideration of the inherent characteristics of SAR targets, which limits the improvement of recognition accuracy and makes poor generalization ability. In addition, due to the black-box characteristics, it is difficult to effectively interpret SAR ATR results. To conquer these problems, we propose an electromagnetic scattering feature (ESF) module embedded network based on attributed scattering center (ASC) model to incorporate the SAR targets' characteristics into the deep learning framework. First, a novel convolutional neural network (CNN)-based algorithm for extracting ASC parameters is proposed, which makes the network focus on target features under the guidance of physical model. Then, the ESF module is designed based on a well-trained ASC parameters extractor to inject the learned target features into the classification network for more robust and interpretable results. Besides, two structures are proposed combined with the ESF module for single-view and multiview SAR target classification, which further illustrates the portability of the module. Experiments on the Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset show the validity of the proposed CNN-based ASC extractor and the ESF module embedded classification network. Compared with ordinary networks, our method can achieve higher classification accuracy under complex conditions, which reflects the better generalization performance of the algorithm. Furthermore, through visualization analysis of the classification results, we show the interpretability of the network combined with the electromagnetic scattering characteristics.**

***Index Terms*— Attributed scattering center (ASC) model, electromagnetic scattering features (ESFs), parameter estimation, physics-guided deep learning, synthetic aperture radar (SAR) target classification.**

I. Introduction

SYNTHETIC aperture radar (SAR) has strong applicability in both military and civilian fields with the advantages of all-time, all-weather, long-range, and high-resolution imaging ability [1]. Spurred by the advantages of SAR, it is widely used in the field of remote sensing image target detection [2], [3], [4], target recognition [5], [6], terrain classification [7], [8], change detection [9], and so on. Automatic target recognition (ATR) is one of the core processes to enhance the intelligent interpretation of SAR images. ATR in these data domains has tried to use various methods, which are either derived from the field of computer vision or specially designed for SAR data, in order to maximize the performance of target recognition [10].

In recent years, deep learning technology applied in the field of computer vision has made great progress [11]. Many scholars apply deep learning algorithms to the field of SAR target classification. Convolutional neural network (CNN) [12] is a classical structure in computer vision, which can learn complex features from the original image with the convolution layer and pool layer. Chen *et al.* [13] first proposed a fully convolutional network (A-ConvNet) for SAR vehicle target classification, which replaces all the fully connected layers with convolutional layers. VGGNet [14], ResNet [15], and DenseNet [16] are the main CNN architectures, which are often used as backbone structures in SAR image interpretation methods based on deep learning [17]. In addition to CNN, recurrent neural networks (RNNs) [18] are another major deep learning network structure. Long short-term memory (LSTM) [19] is one of the most popular architectures in RNN. This structure is widely used in the research of multiview SAR images [20] and has achieved high classification accuracy on sequence vehicle targets. Based on the existing deep learning networks, most of the latest SAR target recognition research is to directly introduce the modules that perform well in the optical image into the original network structure. Wang *et al.* [21] replaced some convolution layers with deformable convolution structures, which improved the classification ability of the algorithm under limited training samples. In [22], a convolutional block attention module (CBAM) [23] was added after each of the convolutional layer to improve the classification performance and efficiency. Lin *et al.* [24] added squeeze and excitation mechanism to the faster R-CNN to further improve the SAR ship detection performance. In [25], the convolutional highway unit was used to train networks with limited SAR data based on highway network architecture, which is

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designed for optical images. Ma *et al.* [26] proposed the method developed from the generative adversarial network (GAN) to solve the open set recognition problem. This type of modification of the network structure can improve the classification accuracy, but it does not combine the specific task characteristics and SAR data electromagnetic features, which makes the network lack interpretability and generalization. Through the above analysis, it can be found that most of the previous studies directly optimized the models generally applied to natural scenes, without considering the different imaging characteristics between SAR images and optical images. As a result, when facing complex experimental conditions in SAR ATR, such as the change of target scene and the decline of data quantity or quality, the robustness of the model is poor.

Different from the deep learning-based method, the traditional target classification algorithm specially designed for SAR data can better reflect the characteristics of SAR targets. Some commonly used SAR target features include geometric features [27], transform domain features [28], and electromagnetic scattering features (ESFs) [29]. Among the traditional methods, the target classification algorithm based on ESFs is a research focus. A theoretical analysis shows that the high-frequency scattering response of the target can be well modeled as the sum of the response of multiple attributed scattering centers (ASCs). Thus, the ASC model can well reflect the physical characteristics of the SAR target. Fu *et al.* [30] proposed a novel template matching-based aircraft recognition method with scattering structure feature (SSF) to improve the classification accuracy in SAR images. Chiang *et al.* [31] proposed a Bayesian method to establish a one-to-one correspondence between two sets of ASCs and used *a posteriori* probability to evaluate their similarity. Ding *et al.* [32] proposed a hierarchical fusion method of the global and local features for SAR ATR. Sparse representation-based classification (SRC) is used to process global features, and local descriptors are calculated by ASC. The results show that the classification under complex experimental conditions depends more on the target electromagnetic scattering characteristics. In addition to scattering features, target-related knowledge, including context knowledge and domain knowledge, can also play a significant role in target recognition. The traditional target recognition method, which closely combines the physical characteristics of the target, makes the classification result interpretable and robust.

Through the analysis and comparison of the above two kinds of methods, we believe that the performance of deep learning-based SAR image target recognition algorithm can be further improved by combining ESFs and target-related knowledge. Kang *et al.* [33] took the surrounding pixels of the target as context knowledge and flattened them to a vector in the fully connected layers of the proposed method, which fused the context information into the network. Also, combined with the context information, Sun *et al.* [34] redesigned the target classification and boundary box regression methods based on the knowledge of target region and background. Zhang *et al.* [35] introduced the domain knowledge of SAR target to assist the target augmentation task for few-shot recognition. Azimuth angle and phase data, which are unique to SAR, are used to enhance target characteristics. Although all of the above methods combine SAR-specific knowledge, there is still a lack of interpretability of the deep learning classification results. Recently, methods that combine the electromagnetic scattering characteristics of the target can be more used to highlight the role of the target scattering structure in the algorithm and provide assistance for algorithm interpretability.

A. Related Work

Considering the advantages of electromagnetic scattering characteristics, researchers began to study how to combine the SAR target features with the deep learning framework to improve the stability and interpretability of the classification results. According to the existing literature, the fusion strategy is a common solution in this field, which can be divided into two main frameworks: feature-level fusion (FLF) [36] and decision-level fusion (DLF) [37]. FLF refers to the combination of features calculated from traditional models with those based on deep learning models. Then, the fused features are subjected to machine learning classifiers to get the final classification results. DLF is the simplest form of fusion between any two or more approaches, which takes the average of the predictions of the deep learning methods and model-based approaches as the final prediction. Cui *et al.* [38] proposed an updated learning method for SAR unlabeled target recognition based on the decision fusion of CNN and other assistant classifiers. Feng *et al.* [39] also presented a decision fusion framework based on target parts divided according to their ASC parameters. Zhang *et al.* [40] proposed a feature fusion framework based on ASC feature and deep learning feature, which fused electromagnetic scattering center features and CNN features into SAR target recognition for the first time. Li *et al.* [41] divided the SAR target into components according to the geometric scattering types of ASCs, and feature fusion is carried out after component features are extracted by the network with an attention mechanism. Kang *et al.* [42] proposed a feature fusion algorithm for SAR target recognition based on a stacked autoencoder (SAE). Ai *et al.* [43] used convolutional kernels of different sizes to extract the multikernel-size deep features of the SAR target and fused them for classification. However, the role of target electromagnetic properties is not well represented by DLF, while complicated feature correlation analysis and fusion operation are required when using FLF due to the differences between features.

In recent research, algorithms based on multitask strategy have been gradually used in target recognition. Sun *et al.* [44] designed a novel few-shot learning framework named scattering characteristics analysis network (SCAN) for SAR aircraft classification. The network uses a scattering extraction module (SEM) to combine the target imaging mechanism with the network. Sun *et al.* [45] also proposed a deep learning-based ship detection and classification network combined with scattering points prediction. These algorithms constrain the training of classification network parameters by taking the learning of target electromagnetic characteristics as an auxiliary task. The specific operation is adding the loss calculation of target

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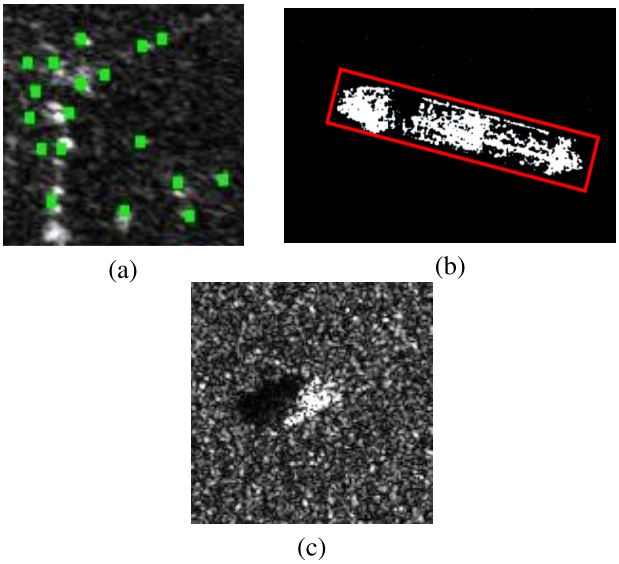


Fig. 1. Scattering characteristics of different targets in SAR images. (a) Aircraft [44]. (b) Ship [45]. (c) Vehicle.

scattering points' distribution to the main classification tasks. In these methods, only a few strong scattering points are considered according to the discrete structural characteristics of aircraft and ship targets, as shown in Fig. 1(a) and (b). However, for the vehicle target in Fig. 1(c), it is difficult to separate isolated scattering points due to the small size of the target. For these small targets, the ASC model can better reflect the scattering characteristics in SAR images. Besides, although these algorithms introduce the target electromagnetic scattering characteristics into the deep learning network, the operation of multitask learning increases the computational complexity of the algorithms. At the same time, through the experiments, we find that the scattering points learning task may affect the performance of classification task.

To introduce the electromagnetic characteristics of SAR target into the deep learning network more conveniently, some researchers designed an algorithm framework of physical model-guided deep learning, which applies the idea of physics-guided neural network (PGNN) [46] and physics-informed neural network (PINN) [47] to the SAR field. Huang *et al.* [48] proposed a novel physics-guided and injected neural network for SAR image classification, which injects the physical scattering characteristics of SAR into the deep neural network. In this framework, deep learning algorithms can directly learn the knowledge of SAR field, and the integration of network layer also makes the fusion structure more reasonable, which inspires the work of this article.

B. Contributions

In order to confront the above issues and improve the stability of target recognition results under complex conditions, an ESF module embedded network structure based on the ASC model is proposed. Unlike the multitask strategy, the algorithm uses a multistage learning strategy, which reduces the computational complexity in the classification process. In the first stage, a deep learning algorithm is used to estimate the target ASC parameters so that the network learns the target electromagnetic scattering characteristics. This step can be performed offline, thus alleviating the computational cost in classification network learning. Then, in the second stage, the network layer feature fusion strategy is adopted to integrate the target electromagnetic characteristics into the recognition network, avoiding the difference between traditional features and deep learning features. Compared with recent SAR target classification methods, the main contributions of this article are given as follows.

1) An intelligent algorithm for parameter estimation of ASC model based on CNN is proposed, which uses the physical model reflecting the scattering characteristics of the target to guide the training process so that the network can focus on the target-related features. It is worth noting that this is the first time to use a deep learning network for ASC model parameter estimation. Experiments show the effectiveness of our algorithm.
2) To enhance the representation of SAR target characteristics in recognition, the framework of ESF module embedded network is constructed. On this basis, two structures are designed for different application scenarios (single-view or multiview images) of classification tasks. The performance of the proposed network structures is verified by experiments. Especially under complex conditions, compared with ordinary networks, the recognition rates of our algorithms are significantly improved.
3) In this article, the complementarity of deep learning network features and electromagnetic scattering characteristics is illustrated by using the feature visualization method for the first time. In addition, combined with the misidentified samples, a preliminary interpretability analysis of the network identification results is conducted.

The remainder of this article is organized as follows. In Section II, the structure of the proposed method and the details of each step are given. Section III presents the experimental results of our method compared with recent algorithms and makes a detailed analysis of the role of the electromagnetic feature module in classification. Finally, Section IV concludes this article.

II. ESF Module Embedded Network

The overall structure of the proposed ESF module embedded network is shown in Fig. 2. The framework consists of three main components: electromagnetic feature extraction, physics-guided deep learning, and network layer embedding. The details of these three modules are described in the following.

A. Electromagnetic Feature Extraction Through ASC Model

The ASC model is motivated by both the physical optics and the geometric theory of diffraction (GTD) [49] and is widely used in the target recognition field. The model assumes that the backscattering of a target can be well approximated as a sum of responses from individual scattering centers as given in the following:

$$E(f, \varphi; \Theta) = \sum_{i=1}^{q} E_i(f, \varphi; \theta_i) \tag{1}$$

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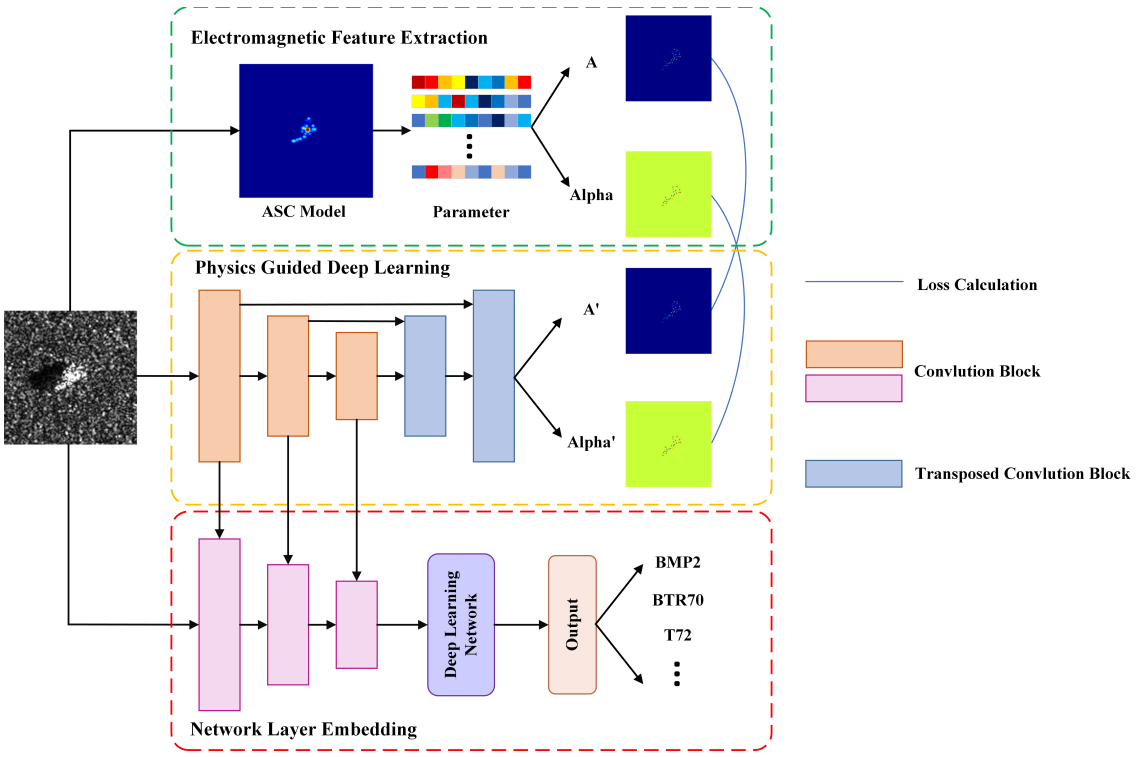


Fig. 2. Overall framework of the proposed ESF module embedded network for SAR target recognition.

where q is the total number of individual scattering centers. For an individual ASC, the backscattered field is parameterized as a function of frequency f and aspect φ

$$E_i(f,\varphi;\theta_i) = A_i \cdot \left(j\frac{f}{f_c}\right)^{\alpha_i} \cdot \exp\left[-j\frac{4\pi f}{C}(x_i\cos\varphi + y_i\sin\varphi)\right] \cdot \operatorname{sinc}\left(\frac{2\pi f}{C}L_i\sin(\varphi-\bar{\varphi}_i)\right) \cdot \exp(-2\pi f\gamma_i\sin\varphi) \quad (2)$$

where f_c is the radar center frequency and C is the propagation velocity of electromagnetic wave. $\Theta = \{\theta_i = [A_i, \alpha_i, x_i, y_i, L_i, \bar{\varphi}_i, \gamma_i]\}(i = 1, 2, \ldots, q)$ is the parameter set, in which (x_i, y_i) are the spatial locations, A_i is the relative amplitude, α_i denotes the frequency and geometry dependence, L_i and $\bar{\varphi}_i$ represent the length and orientation of the distributed ASC, respectively, and γ_i is the aspect dependence of the localized ASC. The details of each parameter are described in [50]. The seven parameters in the ASC model can be predicted by many parameter optimization algorithms, such as genetic algorithm (GA) [51] and approximate maximum likelihood (AML) algorithm [52].

However, the classical parameter optimization algorithm has the problem of low computational efficiency. Deep learning method is a favorable tool to solve such problems. Finding the mapping between images and parameters can be regarded as a regression problem [53], but the number of scattering centers calculated by parameter estimation methods is usually uncertain, which makes the deep learning-based regression model difficult to establish. ASC parameter extraction results are usually saved as parameter matrices. In order to facilitate the construction of subsequent parameter extraction network, we convert the parameter matrix into image form. In this way, we transform the regression problem into an image-to-image calculation problem, which can be solved by using the CNN

$$\text{Image}_A(m,n) = \begin{cases} A_i, & \begin{matrix} m = m_{\text{center}} + x_i \\ n = n_{\text{center}} + y_i \\ (i = 1, 2, \ldots, p) \end{matrix} \\ 0, & \text{others} \end{cases} \quad (3)$$

$$\text{Image}_\alpha(m,n) = \begin{cases} \alpha_i, & \begin{matrix} m = m_{\text{center}} + x_i \\ n = n_{\text{center}} + y_i \\ (i = 1, 2, \ldots, p) \end{matrix} \\ 0, & \text{others.} \end{cases} \quad (4)$$

Since $\bar{\varphi}_i$ and γ_i extraction results are usually small and unstable during the parameter estimation [32], only the other five parameters $[A_i, \alpha_i, x_i, y_i, L_i](i = 1, 2, \ldots, q)$ are considered in this article. The distributed ASC ($L_i > 0$) with a length is represented by two localized ASCs with the same amplitude located at its two ends according to [52]. Such a representation not only reduces the number of attributes, which simplifies the structure of deep learning network, but also contains the length information by the relative positions of the two localized ASCs ($L_i = 0$). Through the above analysis, we mainly consider four parameters $[A_i, \alpha_i, x_i, y_i](i = 1, 2, \ldots, p)$ for each localized ASC. According to the spatial locations (x_i, y_i) of each ASC, we can visualize the rest two parameters A_i and α_i as two electromagnetic feature images of the target. The specific calculation formulas are shown in (3) and (4).

(x_i, y_i) in the ASC model represents the position of the scattering center relative to the image center, and thus, the point

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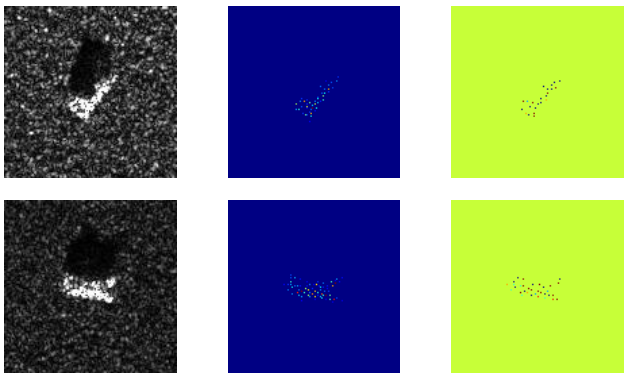


Fig. 3. Visual images of the extracted electromagnetic features. (a) Images of targets. (b) Images of parameter A. (c) Images of parameter α.

coordinate should be added with the center $(m_{\text{center}}, n_{\text{center}})$ during visualization. p represents the updated number of scattering centers after replacing the distributed ASCs with localized ASCs. The ASC parameters of some Moving and Stationary Target Acquisition and Recognition (MSTAR) target images are extracted according to the above process, and the visual images of the extracted electromagnetic features are shown in Fig. 3.

B. Physics-Guided Deep Learning Network for ASC Estimation

The ASC model provides the physical description of target characteristics in the SAR field. Combined with this model, we propose a physics-guided deep learning network for ASC parameters estimation. The structure of the network is shown in Fig. 4.

The network combines the structure of A-ConvNets [13] and FCN [54]. The contracting path (gray blocks) of the network consists of four convolutional layers with 8, 16, 32, and 64 channels of feature maps, and their convolution kernels share the same size of 5×5. Each layer is followed by batch normalization (BN) and a rectified linear unit (ReLU). The max-pooling layer with kernel size 2×2 and stride size 2 is added for downsampling after each convolutional layer. In three branches of expansive path (green blocks), we change the output of the last three convolutional layers of the contracting path into a size of 16×16, 8×8, and 4×4 through multiple convolution operations. The number of channels of the feature maps P is output as the number of ASC parameters needs to be estimated. According to the analysis in Section II-A, we set $P = 2$. Every branch consists of an upsampling operation of the feature map through transposed convolution and then adds to the output of the previous branch. The final output size of the ASC estimation network is $128 \times 128 \times P$.

The size of input images is 128×128 and their corresponding ASC parameter images are used to train the network. A network loss is calculated by the difference between the output images and the parameter images mentioned in Section II-A, as shown in Fig. 2. In this article, the mean square error (MSE) is used as the loss function to evaluate the image difference. The function is to calculate the mean value of the sum of squares of the errors of the corresponding points between the predicted data and the real data as shown in the following equation:

$$\text{MSE}(y, \hat{y}) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{5}$$

where $\hat{y}_i$ means the predicted value of y_i. Then, the prediction loss of parameter images in this article can be written as follows:

$$L_{\text{MSE}} = \frac{1}{2}\text{MSE}\left(\text{image}_A, \widehat{\text{image}_A}\right) + \frac{1}{2}\text{MSE}\left(\text{image}_\alpha, \widehat{\text{image}_\alpha}\right). \tag{6}$$

However, in the process of network training, we find that the above formula can not well describe the difference between the predicted images and the real parameter images in this article. Through the calculation results of the parameter images Fig. 3 shown in Section II-A, it can be found that most values in the images are 0, and nonzero values are only concentrated in a small range of the target area. In this case, the value of MSE calculated by using the whole images is always small, which affects the training of the network. To predict parameters better, we consider calculating the prediction loss separately for the nonzero region of the parameter image as follows:

$$L_{\text{MSE}_{\text{nonzero}}} = \frac{1}{2}\text{MSE}\left(A_{\text{nonzero}}, \hat{A}_{\text{nonzero}}\right) + \frac{1}{2}\text{MSE}(\alpha_{\text{nonzero}}, \hat{\alpha}_{\text{nonzero}}). \tag{7}$$

The final loss function is the sum of MSE of the whole images and MSE of nonzero regions

$$\text{Loss} = L_{\text{MSE}} + L_{\text{MSE}_{\text{nonzero}}}. \tag{8}$$

Under the guidance of physical model, the network can calculate target ASC parameter images automatically. Therefore, we can think that each layer of the network implicitly learns the electromagnetic scattering characteristics of the target. The trained contracting path can be used as the electromagnetic feature extraction module of the target. After obtaining the parameter estimation images, we extract the peak values in $\widehat{\text{Image}_A}$ as the ASC parameter A estimation result and the values in $\widehat{\text{Image}_\alpha}$ corresponding to $\widehat{\text{Image}_A}$'s peak positions as the estimation values of α

$$\begin{cases} \hat{A}_i = \widehat{\text{Image}_A}(m_i, n_i) \\ \hat{\alpha}_i = \widehat{\text{Image}_\alpha}(m_i, n_i) \\ \hat{x}_i = m_i - m_{\text{center}} \\ \hat{y}_i = n_i - n_{\text{center}} \end{cases}, \quad \left\{(m,n)\middle|\widehat{\text{Image}_A}(m,n) = \text{peak}\right\}. \tag{9}$$

Due to the simplification of ASC parameters above, the results of the four parameters estimated by the network are brought into the simplified model of the following formulas for image reconstruction. This simplified model differs from

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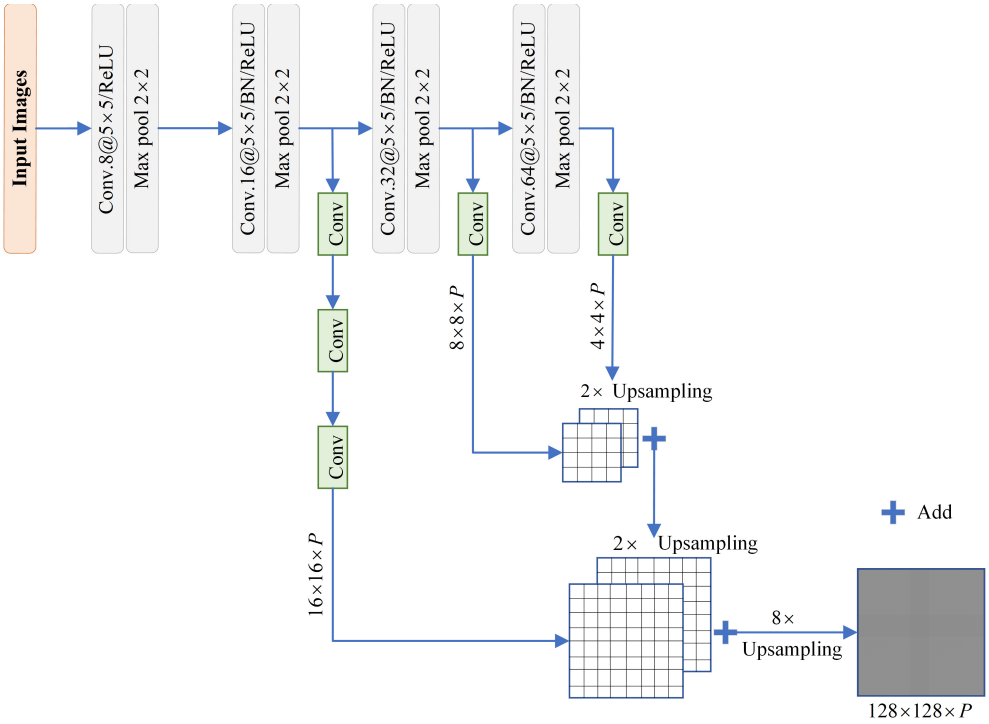


Fig. 4. Structure of the physics-guided deep learning network for ASC parameters estimation.

the ideal point scattering center model in which it still includes target attribute information

$$\hat{E}(f,\varphi;\Theta)=\sum_{i=1}^{\hat{p}}\hat{E}_i(f,\varphi;\theta_i) \tag{10}$$

$$\hat{E}_i(f,\varphi;\theta_i)=\hat{A}_i\cdot\left(j\frac{f}{f_c}\right)^{\hat{\alpha}_i}\cdot\exp\left[-j\frac{4\pi f}{C}(\hat{x}_i\cos\varphi+\hat{y}_i\sin\varphi)\right]. \tag{11}$$

C. ESF Module Embedded Network

After the ASC parameter estimation network is fully trained, the network layer of contracting path is used as the embedded module of target recognition algorithm. We directly add the output of each layer of the contracting path to the corresponding layer in the target recognition network. Different from other common fusion strategies, such as feature fusion and decision fusion, our method performs multilevel fusion at the network layers to avoid the problem of feature domain mismatch between different algorithms. At the same time, the strategy of network layer integration makes our proposed module have good portability and easy to combine with more algorithms. Since the problem of SAR target recognition can be summarized into two application scenarios at this stage: single image recognition and sequence image recognition, the electromagnetic feature extraction module is integrated into two network structures to better illustrate the effectiveness and practicability of the module. Figs. 5 and 6 show the structures of our proposed ESF module embedded in two different SAR image target recognition networks based on deep learning for single image recognition and sequence image recognition, respectively.

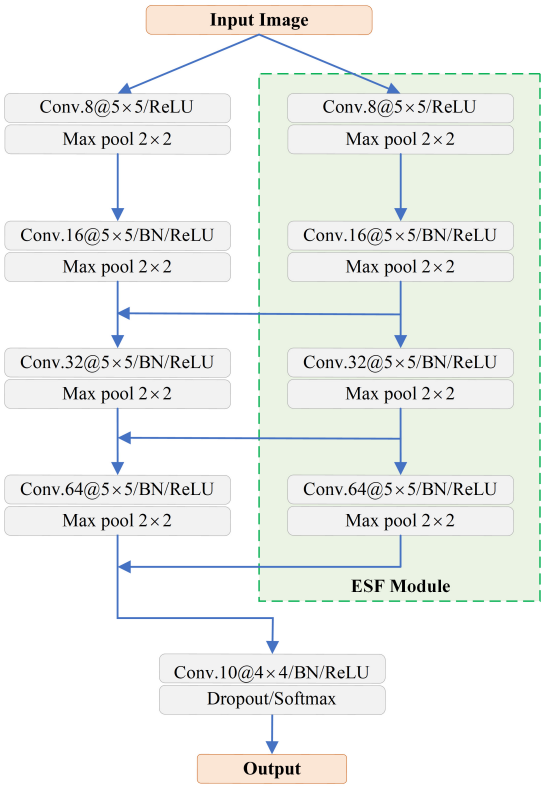


Fig. 5. Structure of ESF module embedded network for single image classification.

In Fig. 5, the ESF module is embedded into the modified A-ConvNets (MACN) [13], which is a classical algorithm in the field of SAR target recognition. In order to embed the proposed electromagnetic feature extraction module into the network, we modify some parameters of A-ConvNets

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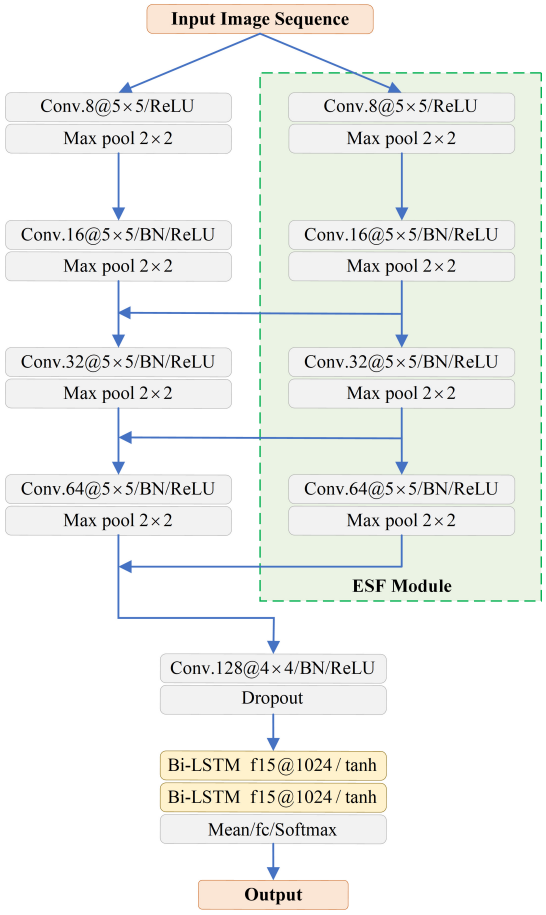


Fig. 6. Structure of ESF module embedded network for sequence image classification.

while maintaining its full convolution characteristics. The structure and parameters of the first four convolution layers are consistent with the proposed module, and the output results of the corresponding layers of the A-ConvNets and the electromagnetic feature extraction module are added together. The output of these four convolution layers is operated by the last convolution layer in Fig. 5 and the final classification probability vector output is obtained through the softmax activation function. In the process of network training, we keep the parameters of the proposed electromagnetic feature extraction module fixed and only train the network of classification algorithm.

In Fig. 6, the ESF module is embedded into the bidirectional convolutional recurrent network (BCRN) [20], which obtains high classification accuracy on the SAR target recognition. The input of the network is an image sequence and the length of the image sequence is 15. For each image in the sequence, the same network structure as that in Fig. 5 is used for feature extraction. For the obtained feature sequence, we use the two-layer Bi-LSTM network structure for learning with 1024 hidden size and tanh activation function. Then, the sequence output is averaged and the final classification result is obtained through full connection and softmax function. The parameters of the proposed electromagnetic feature extraction module are also kept fixed during the training process.

III. Experiments

In order to verify the performance of the proposed method, we first introduce the datasets used in this article, including SAR images for ASC estimation and training and test datasets under different operating conditions for target recognition. In the experimental process, we first evaluate and analyze the performance of the target ASC parameter estimation network. Then, we use the two network structures constructed in Section II-C to test the effect of the proposed ESF module embedded network on the SAR target classification under different conditions. Finally, the classification results of different algorithms are compared and analyzed in detail with visualization methods. All experiments are carried out with both MATLAB platform and the Pytorch deep learning framework on a NVIDIA GeForce RTX 2080 Ti GPU card.

A. Dataset

The measured SAR images used in all of the experiments come from the MSTAR project [55]. This dataset is collected by the Sandia National Laboratory using a Twin Otter SAR sensor operating at the X-band HH polarization. It is taken at multiple depression angles over a full range of viewing angles. The dataset includes two types of target data: ten classes of vehicle targets (BMP2, BTR70, T72, BTR60, 2S1, BRDM2, D7, T62, ZIL131, and ZSU23/4) and one class of a canonical target known as Sandia Laboratory Implementation of Cylinders (SLICY). The optical images of those targets and their corresponding SAR images are shown in Figs. 7 and 8.

For ASC parameter estimation experiment, both vehicle targets data and SLICY data are used to evaluate the parameter estimation results of the network. The SAR images of SLICY were cropped from larger scenes of the measured SAR data. The size of the SLICY target image is 54 × 54 pixels [56]. In order to unify it with the vehicle target image, this article expands SLICY to the size of 128 × 128 and fills the extended area with 0, as shown in Fig. 8.

For target recognition, experiments are conducted under the standard operating condition (SOC) and extended operating conditions (EOCs). The data compositions of vehicle targets data under SOC and EOCs are presented in Tables I–IV.

SOC refers to the training data and test data under similar imaging conditions, using 17° depression angle data of ten kinds of targets as training data and 15° depression angle data as test data. EOC1 corresponds to large depression angle variation, where the training set has 17° depression angle and the test set has 30° depression angle. EOC2/C represents configuration variation of vehicle targets, where training data and test data differ in components. EOC2/V means target version variation, where some structures have relative motion among the training and test sets.

B. Results of ASC Estimation Network

In this section, we use the 17° depression angle vehicle target data under SOC as the input of the ASC parameter estimation algorithm proposed in Section II-B and use the ASC results calculated by the traditional parameter estimation

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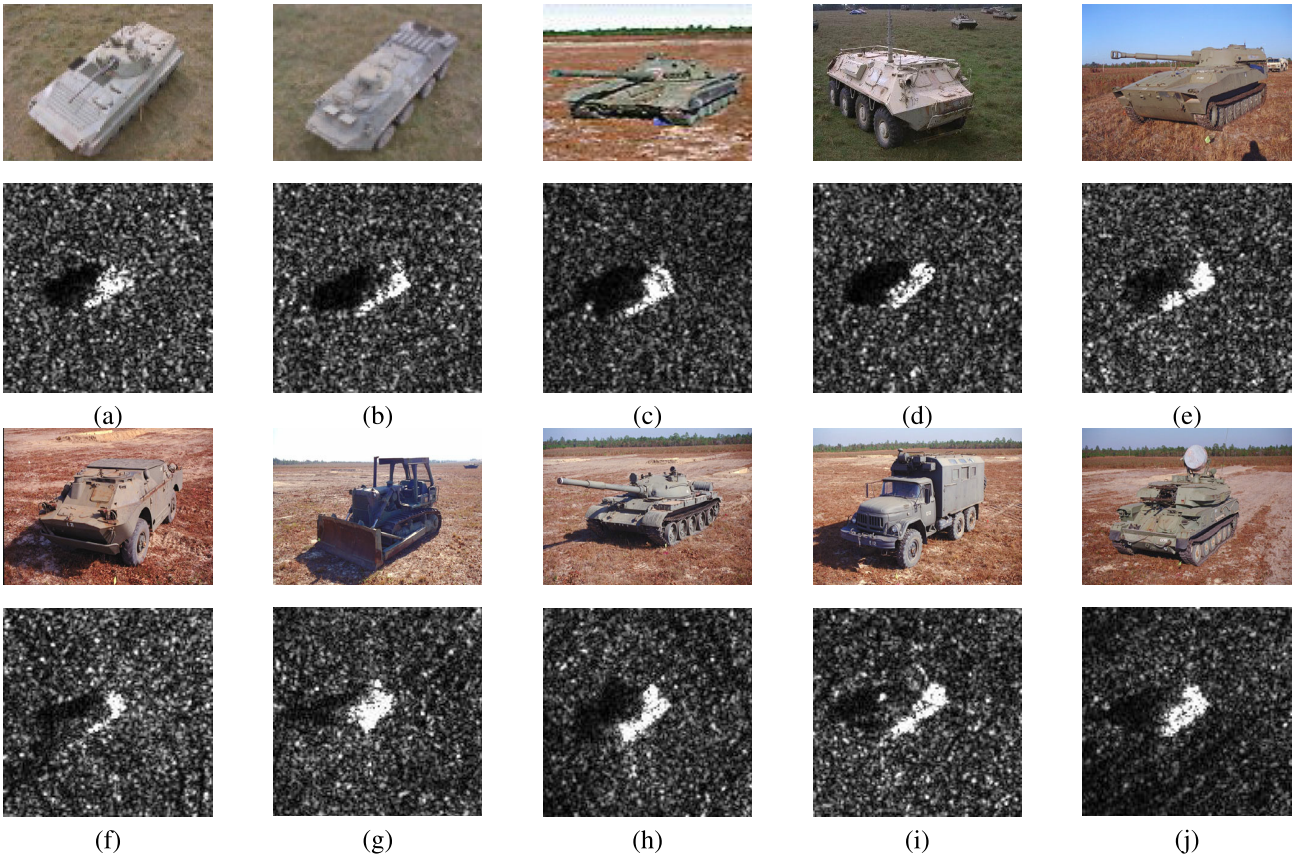


Fig. 7. Optical images of MSTAR vehicle targets and their corresponding SAR images. (a) BMP2. (b) BTR70. (c) T72. (d) BTR60. (e) 2S1. (f) BRDM2. (g) D7. (h) T62. (i) ZIL131. (j) ZSU23/4.

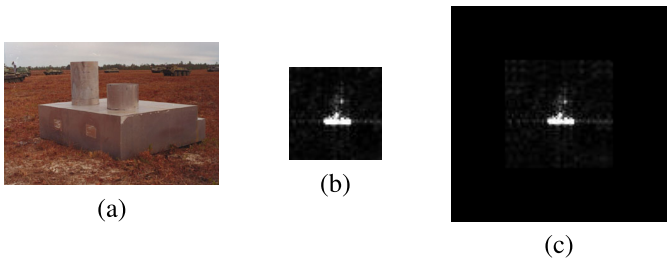


Fig. 8. Optical images of SLICY target and its corresponding SAR image. (a) Optical image. (b) Original SAR image. (c) Expanded SAR image.

TABLE I
DETAILED INFORMATION OF TRAINING AND TEST DATA UNDER SOC

Train			Test		
Type	**Depression**	**Num**	**Type**	**Depression**	**Num**
BMP2	17°	233	BMP2	15°	195
BTR70	17°	233	BTR70	15°	196
T72	17°	232	T72	15°	196
BTR60	17°	256	BTR60	15°	195
2S1	17°	299	2S1	15°	274
BRDM2	17°	298	BRDM2	15°	274
D7	17°	299	D7	15°	274
T62	17°	299	T62	15°	273
ZIL131	17°	299	ZIL131	15°	274
ZSU23/4	17°	299	ZSU23/4	15°	274

TABLE II
DETAILED INFORMATION OF TRAINING AND TEST DATA UNDER EOC1

Train			Test		
Type	**Depression**	**Num**	**Type**	**Depression**	**Num**
T72	17°	299	T72	30°	288
2S1	17°	299	2S1	30°	288
BRDM2	17°	298	BRDM2	30°	287
ZSU23/4	17°	299	ZSU23/4	30°	288

TABLE III
DETAILED INFORMATION OF TRAINING AND TEST DATA UNDER EOC2/C

Train			Test		
Type/Series	**Depression**	**Num**	**Type/Series**	**Depression**	**Num**
BMP2/9563	17°	233	T72/A32	15°, 17°	572
BTR70/C71	17°	233	T72/A62	15°, 17°	573
T72/132	17°	232	T72/A63	15°, 17°	573
BRDM2/E71	17°	298	T72/A64	15°, 17°	573
			T72/S7	15°, 17°	419

method as constraints to train the network. During the test process, both vehicle targets data and SLICY data with 15° depression angle are used to evaluate the ASC parameter estimation results of the network. After obtaining the estimated parameters, we use the image reconstruction method proposed in Section II-B for target simulation. The comparison between the image reconstructed based on the result of traditional parameter estimation method and the image reconstructed

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TABLE IV
DETAILED INFORMATION OF TRAINING AND TEST DATA UNDER EOC2/V

Train			Test		
Type/Series	**Depression**	**Num**	**Type/Series**	**Depression**	**Num**
BMP2/9563	17°	233	BMP2/9566	15°, 17°	428
BTR70/C71	17°	233	BMP2/C21	15°, 17°	429
T72/132	17°	232	T72/A04	15°, 17°	573
BRDM2/E71	17°	298	T72/A05	15°, 17°	573
			T72/A07	15°, 17°	573
			T72/A10	15°, 17°	567
			T72/812	15°, 17°	426

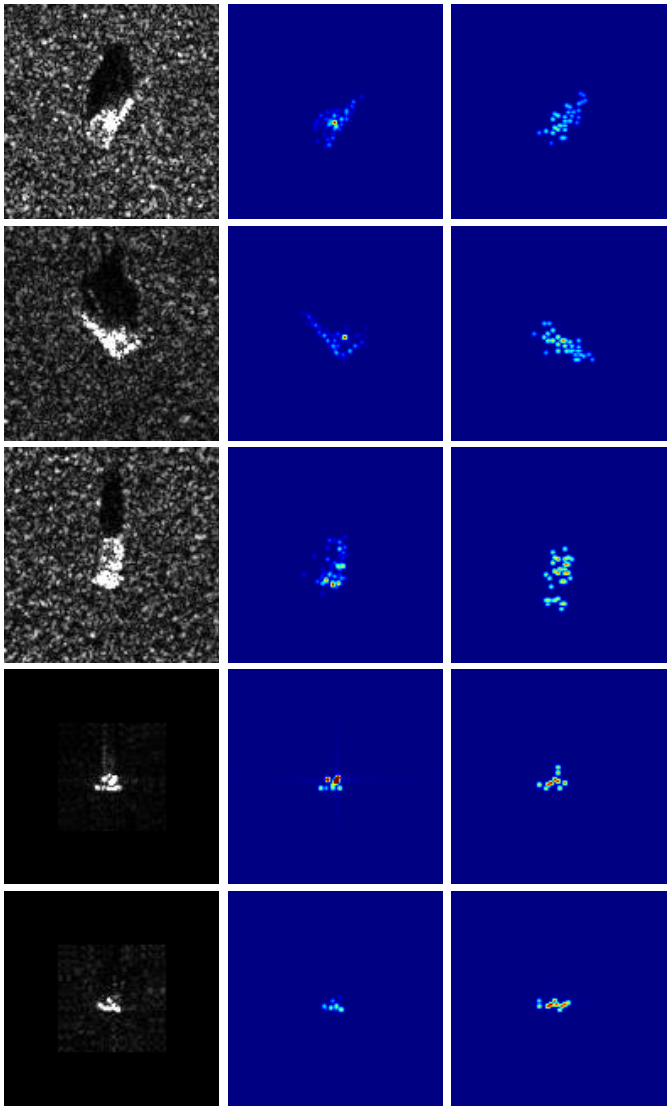


Fig. 9. Image reconstruction results based on traditional parameter estimation method and ASC parameter estimation network. (a) SAR target image. (b) Result based on traditional parameter estimation method. (c) Result based on ASC parameter estimation network.

based on the result of ASC parameter estimation network is shown in Fig. 9.

Compared with the vehicle data reconstruction results of traditional parameter estimation algorithm in the first three rows in Fig. 9, the network output can basically reflect the amplitude and position distribution of scattering points. In addition, comparing the two reconstruction results with the original target image, we find that the reconstructed image based on the ASC parameter estimation network extracts more scattering points than the traditional method, reflecting the more detailed scattering structure of the target. From the results of the second row in Fig. 9, it is easy to find that the deep learning-based method extracts part of the scattering structure information of the low amplitude region of the image target area. However, traditional parameter estimation methods are usually limited by image preprocessing steps such as target segmentation, and the scattering structure of the low amplitude region cannot be obtained. This shows that the deep learning-based method can supplement the shortcomings of the traditional methods to extract the scattering characteristics of the target more completely.

The reconstruction results of SLICY data with different parameter estimation algorithms are shown in the last two rows in Fig. 9. The SLICY target is a precisely designed and machined engineering test target containing standard radar reflector primitive shapes such as flat plates, dihedrals, trihedrals, and top hats [57]. The purpose of this target is to allow image understanding developers the ability to validate the functionality of their algorithm. The target structure is simpler than that of a vehicle, so using SLICY as test data makes it easier to compare the differences in the reconstruction results of the algorithms. The image shows that the ASC parameter estimation network mentioned in this article can learn the location and amplitude information of some target scattering points. However, there are still some errors in estimating the scattering center, and we will continue to optimize the algorithm in the future work.

C. Results of Target Recognition under SOC/EOCs

After the training process of ASC parameter extraction network, we take its contracting path with parameters fixed as an electromagnetic feature extraction module and embed it into different deep learning-based target recognition algorithms. In the target classification experiments, we analyze the performance of the proposed module by comparing the recognition results of the original algorithm and the algorithm embedded with the electromagnetic feature extraction module. In Section II-C, two deep learning algorithms with good effects in the field of SAR target recognition are mentioned: A-ConvNets (ACN) [13] and BCRN [20]. In this article, some parameters are adjusted based on the original algorithms, and the adjusted algorithms are recorded as MACN and modified BCRN (MBCRN). For two novel structures of deep learning algorithms embedded with the electromagnetic feature extraction module as shown in Figs. 5 and 6, we record them as ASC-MACN and ASC-MBCRN, respectively. In addition, we also compare the four structures used in this article with other existing algorithms: VDCNN [58], M-Net [59], and FEC [40]. Table V presents the classification accuracy of different network structures and algorithms under SOC and EOCs.

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TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT NETWORK STRUCTURES UNDER SOC AND EOCS

Method	Accuracy (%)			
	SOC	EOC1	EOC2/C	EOC2/V
ACN [13]	99.13	96.12	98.93	98.60
VDCNN [58]	98.52	94.61	95.46	95.45
M-Net [59]	99.71	97.48	99.67	98.74
FEC [40]	99.59	99.19	-	98.48
BCRN [20]	99.97	99.54	99.81	98.92
MACN	99.46	97.05	99.56	98.09
ASC-MACN	99.42	98.09	99.63	98.07
MBCRN	99.97	99.65	99.94	99.55
ASC-MBCRN	99.96	99.74	99.89	99.64

TABLE VI
MODEL SIZE AND OPERATION TIME OF DIFFERENT STRUCTURES

Method	Model Size (MB)	FPS
MACN	0.30	83.33
ASC-MACN	0.56	83.33
MBCRN	136.12	7.14
ASC-MBCRN	136.39	6.67

It can be found from the results in the table that both modified networks and electromagnetic feature extraction module embedded networks have achieved excellent classification results, which are generally higher than the existing algorithms. Comparing the results of the modified networks and the embedded algorithms, we find that under the condition of sufficient training data, embedding the electromagnetic feature extraction module into the deep learning algorithms to guide their learning process will not significantly affect the classification performance of the algorithms. At the same time, under EOCs, the classification performance of the algorithm is slightly improved after embedding the proposed module.

In addition, we compared the model size and operation time of different structures in Table VI. Frame per second (FPS) is used to measure the time efficiency of different structures. We find that adding the proposed module will not reduce the computational performance of the original algorithm. Considering the classification performance and computational performance, we believe that embedding the electromagnetic feature extraction module in the target classification algorithm will not affect the application of the original algorithm, which means that the module can be quickly applied to different classification tasks based on the existing algorithms.

D. Results of Target Recognition under Complex Conditions

To further analyze the performance of ESF module embedded networks, we conduct experiments under complex conditions in this section.

1) Limited Training Data: First, experiments with limited training data are analyzed. For each network structure, we use 30 samples, 20 samples, and ten samples of each type of SAR targets for network training and then use all the test data in SOC for the target classification experiment. The detailed experimental results are shown in Table VII and Fig. 10 makes an intuitive comparison of the classification performance of each network structure.

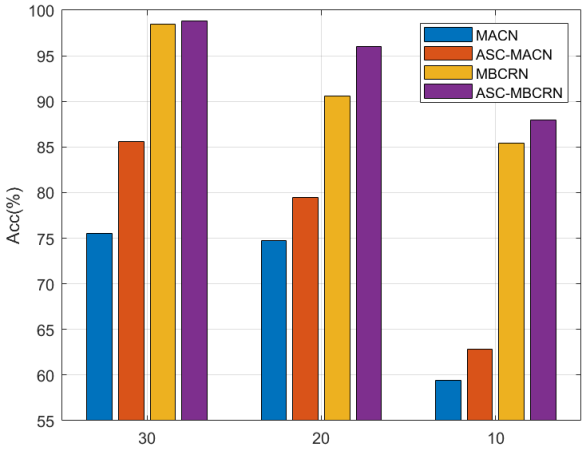


Fig. 10. Classification accuracy of different network structures with limited training data.

TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT NETWORK STRUCTURES WITH LIMITED TRAINING DATA

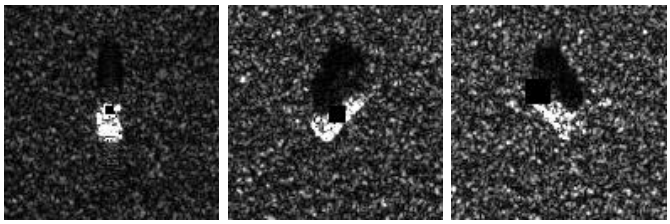
Method	Accuracy (%)			
	SOC	30	20	10
MACN	**99.46**	75.54	74.72	59.38
ASC-MACN	99.42	**85.57**	**79.46**	**62.85**
MBCRN	**99.97**	98.43	90.60	85.40
ASC-MBCRN	99.96	**98.84**	**96.04**	**87.96**

With the decrease in the number of training samples, the classification performance of all network structures decreases significantly, especially the networks without electromagnetic features. In contrast, embedding electromagnetic feature extraction module can significantly improve the classification performance of the algorithm in the case of few shot. This also shows that in the case of insufficient training samples, the electromagnetic feature extraction module can provide new auxiliary information for network training and guide the network for better classification.

2) Random Occlusions: In view of the target masking under the actual observation conditions, we cover part of the target image area to further test the effectiveness of the proposed electromagnetic feature extraction module. Since the target is mainly concentrated in the central area of the slice image, we randomly select a square area in the middle area of the image and set its value to 0 and the range of central region is 64×64. Three different cover sizes are used for experiments: 5×5, 10×10, and 15×15. Fig. 11 shows the target images after random occlusion. The black squares in the target area are the occluded area. All network structures are trained on SOC datasets, and then, the performance of the algorithms is evaluated by using the randomly occluded SOC test set data.

The detailed experimental results are shown in Table VIII. Fig. 12 makes an intuitive comparison of the classification performance of each network structure. When the occlusion

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(a) (b) (c)

Fig. 11. Target images after random occlusion. Cover sizes of (a)–(c) are 5 × 5, 10 × 10, and 15 × 15, respectively.

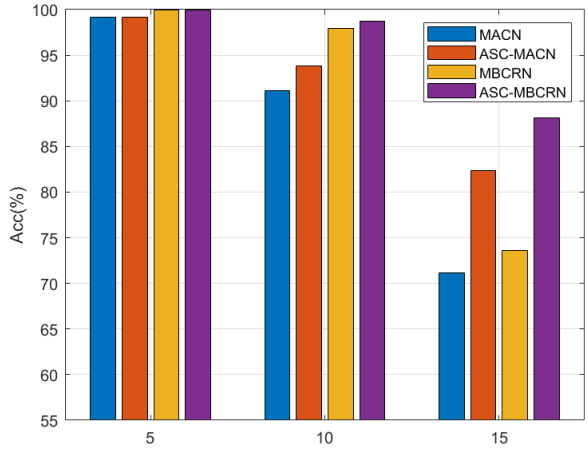


Fig. 12. Classification accuracy of different network structures with randomly occluded test data.

TABLE VIII

CLASSIFICATION ACCURACY OF DIFFERENT NETWORK STRUCTURES WITH RANDOMLY OCCLUDED TEST DATA

Method	Accuracy (%)			
	SOC	5	10	15
MACN	**99.46**	99.18	91.09	71.18
ASC-MACN	99.42	**99.18**	**93.86**	**82.39**
MBCRN	**99.97**	99.97	97.98	73.61
ASC-MBCRN	99.96	**99.96**	**98.76**	**88.16**

area is small, the four network structures can both achieve very high classification accuracy. However, when the size becomes larger, the algorithms embedded with the electromagnetic feature extraction module can achieve more stable classification results compared with others. We believe that after using electromagnetic features to guide network learning, the algorithm can pay more attention to the scattering characteristics of SAR targets and provide additional knowledge for classification tasks. When the target image has random occlusion, the target electromagnetic characteristics learned by the network can make up for the lack of input data quality.

E. Discussion of Experimental Results

In this section, we combine visualization methods to make a more detailed analysis of the proposed module and algorithms. Because the network for a single image learns the characteristics of one sample when classifying, it is easier to observe the role of electromagnetic scattering characteristics in the classification process. Therefore, in this section, we use

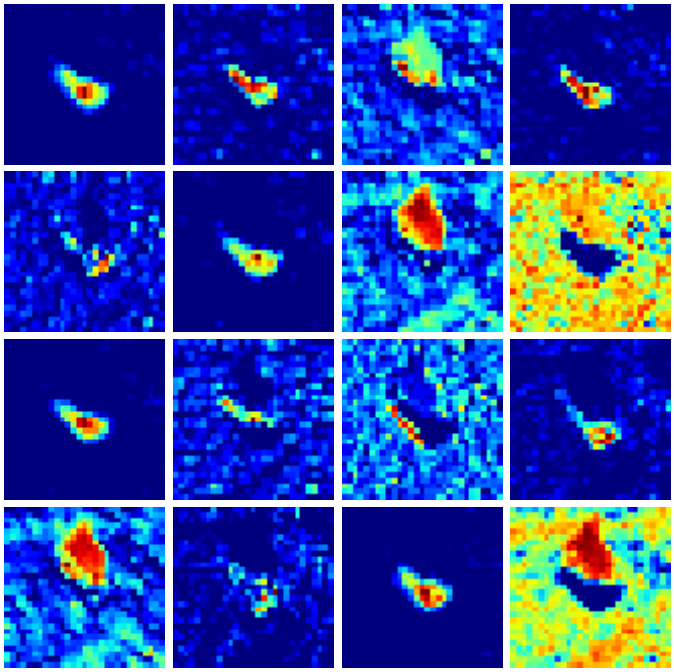


Fig. 13. Feature maps for MACN.

the SOC experimental results of MACN and ASC-MACN to compare and analyze. The layer we visualize is the first network fusion layer in ASC-MACN.

1) Complementarity of Features: To analyze the differences and complementarities between the electromagnetic scattering and the general CNN features, we output the feature maps of the corresponding layers in the MACN and ASC parameter estimation network proposed in Section II-B, as shown in Figs. 13 and 14, respectively.

From Fig. 13, it is easy to see that the common convolution neural network pays attention not only to the target area pixels but also to the background area pixels when classifying SAR images. The learning of the target area is reflected in the recognition of the target as a whole, and the pixel points of interest cover a large area on the target. At the same time, nearly one-third of the channels learn the background and target shadow information from SAR images. However, the feature maps of the ASC parameter estimation network Fig. 14 are obviously more focused on the target area, and only one channel is used to learn the target background information. In addition, the feature maps of the network have more scattered active pixel points in the target area, which can be interpreted as the learning of the target area being more focused on the local area of the target. It is also due to this property that we can find that in contrast to MACN, there is no channel in the ASC parameter estimation network that can learn completely about the global properties of the target, such as the target's profile information.

From the above analysis, we can find that there is complementarity between common CNN features and ESFs: global information learned by CNNs and local information learned by the ESF module. Fig. 15 shows the feature maps of corresponding layer of ASC-MACN, which fuses deep learning network and SAR image electromagnetic scattering properties.

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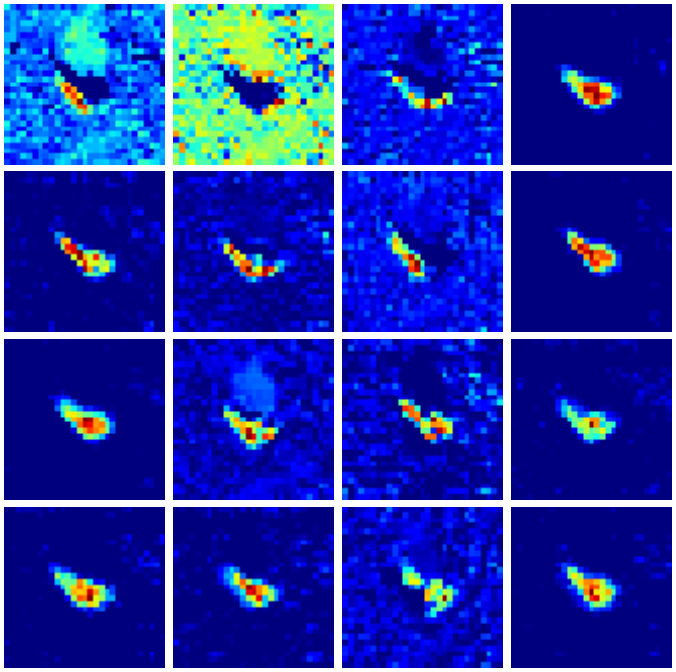


Fig. 14. Feature maps for ASC parameter estimation network.

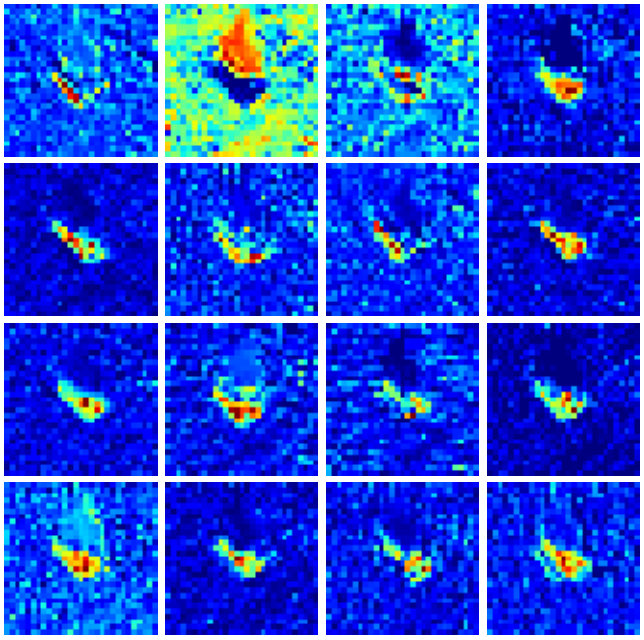
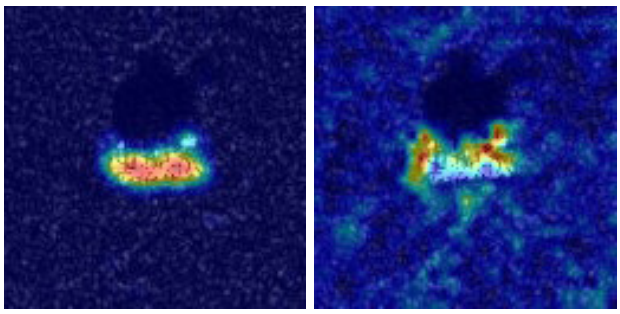


Fig. 15. Feature maps for ASC-MACN.

Integrating the features of above two networks, the graph pays more attention to the local scattering information of the target and also contains the complete global information. However, since the SAR image background also contains scattering information, the signal-to-noise ratio of the target region channels in the ASC-MACN feature maps is low compared to that of MACN. This also explains why, under SOC, the ASC-MACN recognition ability is slightly lower than the original network. This also suggests a direction for future research: how to better embed the features of electromagnetic modules into the original classification network.



(a) (b)

Fig. 16. Visualization with Grad-CAM. (a) MACN. (b) ASC-MACN.

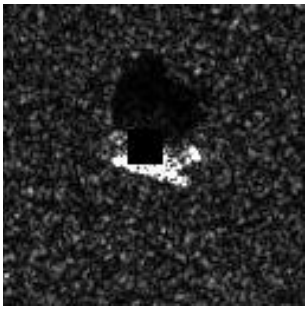


Fig. 17. Occluded sample in test data.

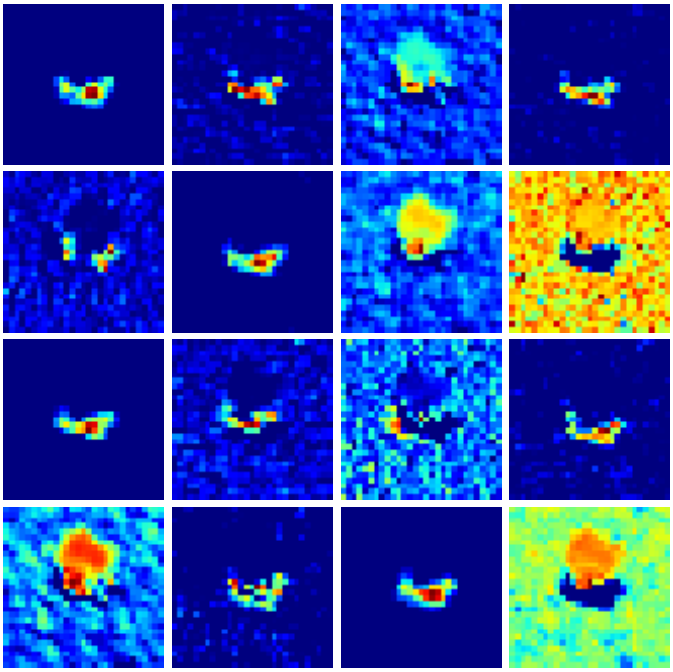


Fig. 18. Feature maps for MACN with the input of occluded sample.

We also use Grad-CAM [60] to visualize MACN and ASC-MACN. The result in Fig. 16 also shows that with the addition of ESF module, the network will focus more on the local area of the target.

2) Analysis of Misclassified Sample: In the previous occlusion experiment, we found that a single image classification network with an ESF module performs even better than the original sequence image classification network. We screen a partially occluded sample shown in Fig. 17 that is misclassified in MACN but correctly classified in ASC-MACN to further

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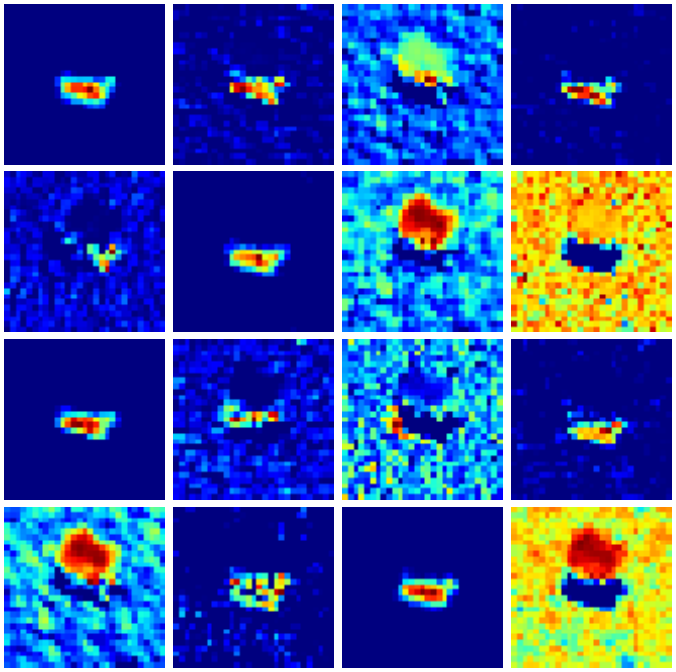


Fig. 19. Feature maps for MACN with the input of original sample.

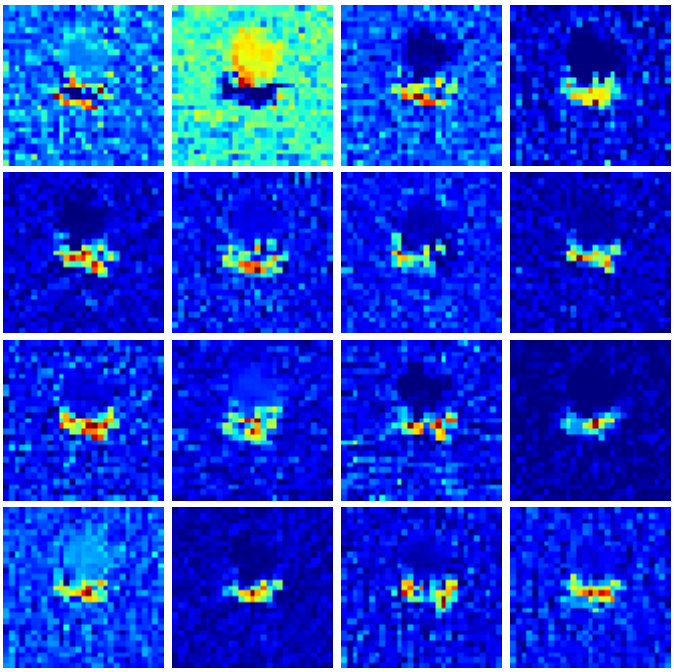


Fig. 20. Feature maps for ASC-MACN with the input of occluded sample.

analyze the role of ESF modules under complex experimental conditions. The original image of the sample is classified correctly with MACN.

Figs. 18 and 19 show the feature maps of the MACN when the occluded image and the original image are input data, respectively. Comparing the two figures, it can be found that when there is occlusion in the target area, the pixel value of the feature image at the corresponding position is also missing. Combined with the previous analysis, we know that an ordinary CNN network learns the global information of the target area as category knowledge. When the target is partially occluded, the global information will change greatly, which will affect the classification results.

Fig. 20 presents the feature maps of ASC-MACN, which identifies the category of the partially occluded sample. From the target area channel and background channel in the feature map, we can still see that occlusion has a certain impact on the feature map. However, unlike MACN, ASC-MACN focuses more on scattered pixels. Therefore, we can infer that due to the combination of ESF module, the network pays more attention to the local information of the target, and the blocked area of the target does not cause the loss of key identification information, so it can be correctly classified by ASC-MACN. This result also shows the effectiveness and robustness of the proposed ESF module embedded network under complex experimental conditions from the perspective of interpretability.

IV. Conclusion

In this article, an ESF module embedded network based on the ASC model is proposed for robust and interpretable SAR ATR. First, the intelligent algorithm for ASC parameter estimation based on CNN is designed, which introduces the physical model into a deep learning network. On this basis, the ESF module is proposed, which can incorporate the electromagnetic scattering characteristics of targets into the target classification network to obtain more stable results. Compared with other existing algorithms, the ESF module embedded network achieves better classification performance, especially in complex experimental conditions, such as limited training samples or occluded test samples. In addition, we also analyze the complementarity between SAR image electromagnetic characteristics and ordinary network characteristics through visualization and further illustrate the rationality of feature fusion algorithm. At the same time, combined with the misclassified samples, the interpretability of the classification results is preliminarily discussed. In future work, we will further optimize the ASC parameter extraction network structure to achieve more accurate results. Based on the visual analysis of the algorithm, we will explore the adaptive fusion method of target electromagnetic features and network features.

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